

PhysicsLab — Barcha Fizika Formulalari (10-11 Sinf)

Fizika fanining barcha asosiy formulalari — O'zbekiston o'rta ta'lim dasturiga asoslangan. Har bir formula KaTeX formatida, o'zbek tilida tushuntirilgan.

MUNDARIJA

- 1. Kinematika (Harakat geometriyasi)
- 2. Dinamika (Harakat dinamikasi)
- 3. Mexanikada saqlanish qonunlari
- 4. Statika va Gidrodinamika
- 5. Mexanik tebranishlar va to'lqinlar
- 6. Termodinamika
- 7. Elektrostatika
- 8. O'zgarmas tok
- 9. Elektromagnit induksiya
- 10. Optika
- 11. Atom va yadro fizikasi
- 12. Maxsus nisbiylik nazariyasi

1. KINEMATIKA

1.1 To'g'ri chiziqli tekis harakat

Formula	Tavsif	Belgilar
$v = \frac{s}{t}$	O'rtacha tezlik	v —tezlik (m/s), s —yo'l (m), t —vaqt (s)
$s = vt$	Bosib o'tilgan yo'l	s —yo'l, v —tezlik, t —vaqt

1.2 To'g'ri chiziqli tekis o'zgaruvchan harakat

Formula	Tavsif	Belgilar
$v = v_0 + at$	Tezlik tenglamasi	v —boshlang'ich tezlik (m/s), a —tezlanish (m/s²)
$s = v_0t + \frac{at^2}{2}$	Yo'l tenglamasi	s —bosib o'tilgan yo'l (m)

Table 2 – continued

Formula	Tavsif	Belgilar
$s = \frac{v_0 + v}{2} \cdot t$	Yo' l (o' rtacha tezlik orqali)	v —oxirgi tezlik (m/s)
$v^2 - v_0^2 = 2as$	Tezlik va yo' l bog' lanishi	Tekis tezlanuvchan harakat
$s = vt - \frac{at^2}{2}$	Sekinlanuvchan harakat uchun	$a < 0$ bo' lganda

1.3 Erkin tushish (Galileo qonunlari)

Formula	Tavsif	Belgilar
$v = gt$	Erkin tushish tezligi	$g = 9.8 \text{ m/s}^2$ —og' irlik kuchi tezlanishi
$h = \frac{gt^2}{2}$	Tushish balandligi	h —balandlik (m)
$v = \sqrt{2gh}$	Tushish tezligi (balandlikdan)	Torrichelli formulisi
$t = \sqrt{\frac{2h}{g}}$	Tushish vaqti	h balandlikdan tushish vaqti

1.4 Aylana bo'ylab harakat

Formula	Tavsif	Belgilar
$\omega = \frac{\varphi}{t}$	Burchak tezligi	ω —burchak tezlik (rad/s), φ —burchak (rad)
$\omega = \omega_0 + \varepsilon t$	Aylanma harakat	ε —burchak tezlanish (rad/s ²)

Table 4 – continued

Formula	Tavsif	Belgilar
$v = \omega R$	Chiziqli tezlik	v —chiziqli tezlik (m/s), R —radius (m)
$a_n = \frac{v^2}{R} = \omega^2 R$	Markazga intilma tezlanish	a —normal tezlanish (m/s ²)
$T = \frac{2\pi R}{v} = \frac{2\pi}{\omega}$	Aylanish davri	T —davr (s)
$n = \frac{1}{T} = \frac{\omega}{2\pi}$	Aylanish chastotasi	n —chastota (Hz)

1.5 Masshtabli harakat (nisbiy harakat)

Formula	Tavsif	Belgilar
$\vec{v}_{rel} = \vec{v}_{abs} - \vec{v}_{sanoq}$	Nisbiy tezlik	Galiley transformatsiyasi

2. DINAMIKA

2.1 Nyuton qonunlari

Formula	Tavsif	Belgilar
$\vec{F} = m\vec{a}$	Nyuton 2-qonuni	F —kuch (N), m —massa (kg), a —tezlanish (m/s ²)
$F_{12} = -F_{21}$	Nyuton 3-qonuni	Bir jismga ta' sir = ikkinchiga teskari ta' sir
$\vec{F}_{res} = \sum \vec{F}_i = 0$	Nyuton 1-qonuni (muvozanat)	Tekis harakat yoki tinchlik

2.2 Gravitatsiya

Formula	Tavsif	Belgilar
$F = G \frac{m_1 m_2}{r^2}$	Univers gravitatsiya qonuni	$G = 6.67 \times 10^{-11} \text{ N}\cdot\text{m}^2/\text{kg}^2$ – gravitatsion doimiy
$g = G \frac{M}{R^2}$	Erkin tushish tezlanishi	M –Yer massasi, R –Yer radiusi
$P = mg$	Jism og'irligi	P –og'irlilik (N), m –massa (kg)
$g_h = G \frac{M}{(R + h)^2}$	Balandlikdagi g	h –Yer yuzasidan balandlik

2.3 Ishtirok kuchi va kuch momenti

Formula	Tavsif	Belgilar
$F_{elan} = kx$	Guk qonuni	k –elastiklik koeffitsienti (N/m), x – uzayish (m)
$M = F \cdot l$	Kuch momenti	M –moment (N·m), l –yelka (m)
$F_{tren} = \mu N$	Ishtirok kuchi	–ishtirok koeffitsienti, N –normal reaksiya (N)

2.4 Kuch ta'sirida harakat (maxsus holatlar)

Formula	Tavsif	Belgilar
$a = g(\sin \alpha - \mu \cos \alpha)$	Qiya tekislikda harakat	–qiyalik burchagi
$F_{cent} = m\omega^2 R = \frac{mv^2}{R}$	Markazga intilma kuch	Aylana bo'ylab harakat

3. MEXANIKADA SAQLANISH QONUNLARI

3.1 Impuls va impuls saqlanishi

Formula	Tavsif	Belgilar
$\vec{p} = m\vec{v}$	Mexanik impuls	p —impuls (kg·m/s)
$\vec{F} \cdot \Delta t = \Delta \vec{p}$	Impuls o' zgarishi	F·Δt —kuch impulsi
$m_1\vec{v}_1 + m_2\vec{v}_2 = m_1\vec{v}'_1 + m_2\vec{v}'_2$	Impuls saqlanish qonuni	To' qnashishda

3.2 Mexanik ish va energiya

Formula	Tavsif	Belgilar
$A = Fs \cos \alpha$	Mexanik ish	A —ish (J), —kuch va yo' l orasidagi burchak
$E_k = \frac{mv^2}{2}$	Kinetik energiya	E —kinetik energiya (J)
$E_p = mgh$	Potensial energiya	E —potensial energiya (J), h — balandlik (m)
$E_{pr} = \frac{kx^2}{2}$	Prujina potensial energiyasi	k —elastiklik koeffitsienti
$E_{total} = E_k + E_p = const$	Energiya saqlanish qonuni	Mexanik energiya saqlanadi (konservativ kuchlar uchun)
$N = \frac{A}{t}$	Quvvat	N —quvvat (Vt)
$N = Fv \cos \alpha$	Quvvat (kuch va tezlik orqali)	—

Table 11 – continued

Formula	Tavsif	Belgilar
$\eta = \frac{A_{foydali}}{A_{umumiy}} \cdot 100\%$	F.I.K. (foydali ish koeffitsienti)	–KPD (%)

4. STATIKA VA GIDRODINAMIKA

4.1 Suyuqlik va gaz bosimi

Formula	Tavsif	Belgilar
$p = \frac{F}{S}$	Bosim	p –bosim (Pa), S –yuza (m ²)
$p = \rho gh$	Gidrostatik bosim	–zichlik (kg/m ³), h –chuqurlik (m)
$p = p_0 + \rho gh$	Absolyut bosim	p –atmosfera bosimi (10 Pa)

4.2 Arkhimed qonuni

Formula	Tavsif	Belgilar
$F_A = \rho_{suyuq} g V_{tush}$	Ko' taruvchi kuch	V –tushgan hajm (m ³)

4.3 Suyuqlik oqimi

Formula	Tavsif	Belgilar
$S_1 v_1 = S_2 v_2$	Uzluksizlik tenglamasi	S –kesim yuzasi (m ²), v –oqim tezligi (m/s)
$p + \frac{\rho v^2}{2} + \rho gh = const$	Bernulli tenglamasi	Gidrodinamikada energiya saqlanishi

5. MEXANIK TEBRANISHLAR VA TO'LQINLAR

5.1 Garmonik tebranishlar

Formula	Tavsif	Belgilar
$x = A \sin(\omega t + \varphi_0)$	Tebranish tenglamasi	A —amplituda (m), φ_0 —burchak chastota (rad/s), t —boshlang' ich faza
$T = \frac{2\pi}{\omega} = \frac{1}{f}$	Davr	T —davr (s), f —chastota (Hz)
$\omega = 2\pi f$	Burchak chastota	—
$v = A\omega \cos(\omega t + \varphi_0)$	Tezlik	Garmonik tebranishda
$a = -A\omega^2 \sin(\omega t + \varphi_0)$	Tezlanish	$a = -\omega^2 x$
$v_{max} = A\omega$	Maksimal tezlik	Muvozanat nuqtasida
$a_{max} = A\omega^2$	Maksimal tezlanish	Ekstremal nuqtalarda

5.2 Prujina va mayatnik

Formula	Tavsif	Belgilar
$T = 2\pi \sqrt{\frac{m}{k}}$	Prujina-mass davri	k —prujina koeffitsienti (N/m)
$T = 2\pi \sqrt{\frac{l}{g}}$	Matematik mayatnik davri	l —uzunlik (m), g = 9.8 m/s ²
$\omega = \sqrt{\frac{k}{m}}$	Prujina burchak chastotasi	—

Table 16 – continued

Formula	Tavsif	Belgilar
$\omega = \sqrt{\frac{g}{l}}$	Mayatnik burchak chastotasi	—

5.3 Tebranish energiyasi

Formula	Tavsif	Belgilar
$E = \frac{kA^2}{2} = \frac{m\omega^2 A^2}{2}$	Jami energiya	Garmonik tebranishda energiya saqlanadi
$E_k = \frac{mv^2}{2} = \frac{m\omega^2(A^2 - x^2)}{2}$	Kinetik energiya	x nuqtadagi kinetik energiya
$E_p = \frac{kx^2}{2}$	Potensial energiya	x siljishdagi potensial energiya

5.4 To'liqlar

Formula	Tavsif	Belgilar
$v = \lambda f = \frac{\lambda}{T}$	To'liqlar tezligi	—to'liqlar uzunligi (m), v —tezlik (m/s)
$\lambda = vT = \frac{v}{f}$	To'liqlar uzunligi	—
$k = \frac{2\pi}{\lambda}$	To'liqlar soni	k —vektor (rad/m)
$\omega = 2\pi f$	Burchak chastota	To'liqlar uchun
$y = A \sin(\omega t - kx)$	Yoyiluvchi to'liqlar tenglamasi	x nuqtada, t vaqtda siljish

Table 18 – continued

Formula	Tavsif	Belgilar
$y = 2A \cos(kx) \sin(\omega t)$	To'xtama to'lqin	Ikki to'lqin superpozitsiyasi

5.5 To'lqinlar (maxsus holatlar)

Formula	Tavsif	Belgilar
$v = \sqrt{\frac{F}{\mu}}$	Sirg'anuvchi to'lqin (tor)	F —cho'zish kuchi (N), —chiziqli zichlik (kg/m)
$v = \sqrt{\frac{E}{\rho}}$	Uzuvchanlik to'lqin	E —Yunq moduli (Pa), —zichlik
$I = \frac{P}{S}$	To'lqin intensivligi	I —intensivlik (Vt/m ²), P —quvvat (Vt)

6. TERMODINAMIKA

6.1 Gaz qonunlari

Formula	Tavsif	Belgilar
$\frac{p_1 V_1}{T_1} = \frac{p_2 V_2}{T_2}$	Kombinirlangan gaz qonuni	Ideal gaz uchun
$pV = \frac{m}{M}RT = \nu RT$	Mendelev-Klapeyron tenglamasi	M —molyar massa (kg/mol), R = 8.31 J/(mol·K)
$p = nkT$	Bosim (molekulyar nazariya)	n —konsentratsiya (m ⁻³), k = 1.38 × 10 ⁻²³ J/K —Boltsman doimiyasi
$pV = NkT$	Ideal gaz (zarracha soni)	N —zarracha soni

6.2 Gaz jarayonlari

Formula	Tavsif	Belgilar
$pV = const$	Izotermik jarayon	$T = const$ (Boyl-Mariott qonuni)
$\frac{V}{T} = const$	Izobarik jarayon	$p = const$ (Gey-Lyussak qonuni)
$\frac{p}{T} = const$	Izoxorik jarayon	$V = const$ (Sarl qonuni)
$pV^\gamma = const$	Adiabatik jarayon	$\gamma = C_p / C_v$ —adiabatik ko'rsatkich
$TV^{\gamma-1} = const$	Adiabatik (2-shakl)	—
$\frac{p^{\gamma-1}}{T^\gamma} = const$	Adiabatik (3-shakl)	—

6.3 Issiqlik miqdori

Formula	Tavsif	Belgilar
$Q = cm\Delta T$	Issiqlik miqdori	c —issiqlik sig'imi ($J/(kg \cdot K)$)
$Q = \lambda m$	Erish issiqligi	λ —yashirin issiqlik (J/kg)
$Q = rm$	Bug'lanish issiqligi	r —bug'lanish issiqligi (J/kg)
$Q = qm$	Yoqilg'i issiqligi	q —yonish issiqligi (J/kg)

6.4 Termodinamika qonunlari

Formula	Tavsif	Belgilar
$\Delta U = Q - A$	Termodinamika 1-qonuni	ΔU —ichki energiya o' zgarishi (J)
$A = p\Delta V$	Gazning kengayish ishi	A —ish (J)
$A = \nu RT \ln \frac{V_2}{V_1}$	Izotermik jarayon ishi	—
$\Delta U = \frac{i}{2} \nu R \Delta T$	Ichki energiya o' zgarishi	i —erkinlik darajasi
$\eta = \frac{A_{foydali}}{Q_{olgan}} \cdot 100\%$	F.I.K. (KPD)	—foydali ish koeffitsienti (%)
$\eta_{Karno} = \frac{T_1 - T_2}{T_1} \cdot 100\%$	Karno siklining KPD	T —qizdirgich temperaturasi (K), T —sovitgich temperaturasi (K)
$\eta = \frac{Q_1 - Q_2}{Q_1} \cdot 100\%$	KPD (issiqlik orqali)	Q —olingan issiqlik, Q —berilgan issiqlik

6.5 Molekulyar-kinetik nazariya

Formula	Tavsif	Belgilar
$\frac{m_0 v^2}{2} = \frac{3}{2} kT$	Molekula o' rtacha kinetik energiyasi	m —molekula massasi (kg)
$v_{kv} = \sqrt{\frac{3kT}{m_0}}$	Kvadratik o' rtacha tezlik	—
$v_{kv} = \sqrt{\frac{3RT}{M}}$	Kvadratik o' rtacha (molyar)	M —molyar massa
$U = \frac{i}{2} \nu RT = \frac{i}{2} NkT$	Ichki energiya	i —erkinlik darajasi

Table 24 – continued

Formula	Tavsif	Belgilar
$C_V = \frac{i}{2}R$	Molyar issiqlik sig'imi (V=const)	C –izoxorik issiqlik sig'imi (J/(mol·K))
$C_p = C_V + R = \frac{i+2}{2}R$	Molyar issiqlik sig'imi (p=const)	C –izobarik issiqlik sig'imi
$\gamma = \frac{C_p}{C_V} = \frac{i+2}{i}$	Adiabatik ko'rsatkich	–(gamma)

7. ELEKTROSTATIKA

7.1 Koulon qonuni va elektr maydon

Formula	Tavsif	Belgilar
$F = k \frac{q_1 q_2}{r^2} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2}$	Koulon qonuni	$k = 9 \times 10^9 \text{ N}\cdot\text{m}^2/\text{C}^2$, $\epsilon_0 = 8.85 \times 10^{-12} \text{ F/m}$
$E = \frac{F}{q}$	Elektr maydon kuchlanishi	E –kuchlanish (N/C yoki V/m)
$E = k \frac{q}{r^2}$	Nuqtaviy zaryad maydoni	–
$\vec{E} = \vec{E}_1 + \vec{E}_2 + \dots$	Superpozitsiya prinsipi	Maydon vektorlari qo'shiladi

7.2 Potensial va ish

Formula	Tavsif	Belgilar
$\varphi = \frac{E_p}{q}$	Potensial	–potensial (V)
$\varphi = k \frac{q}{r}$	Nuqtaviy zaryad potentsiali	–

Table 26 – continued

Formula	Tavsif	Belgilar
$A = q(\varphi_1 - \varphi_2) = qU$	Elektr maydonda ish	U –kuchlanish (V)
$U = \varphi_1 - \varphi_2$	Kuchlanish (potensial farqi)	U –volt (V)
$E = \frac{U}{d}$	Bir jinsli maydon kuchlanishi	d –masofa (m)

7.3 Kondensator va sig'im

Formula	Tavsif	Belgilar
$C = \frac{q}{U}$	Sig' im	C –farada (F)
$C = \frac{\varepsilon_0 \varepsilon S}{d}$	Tekis kondensator sig' imi	–dielektrik o' tkazuvchanligi, S – yuza (m ²)
$W = \frac{qU}{2} = \frac{CU^2}{2} = \frac{q^2}{2C}$	Kondensator energiyasi	W –energiya (J)
$C = C_1 + C_2 + \dots$	Parallel ulash	Sig' imlar qo' shiladi
$\frac{1}{C} = \frac{1}{C_1} + \frac{1}{C_2} + \dots$	Ketma-ket ulash	Teskari sig' imlar qo' shiladi

8. O'ZGARMAS TOK

8.1 Asosiy qonunlar

Formula	Tavsif	Belgilar
$I = \frac{q}{t}$	Tok kuchi	I —amper (A), q —zaryad (C)
$I = \frac{\Delta q}{\Delta t}$	Tok kuchi (ixtiyoriy)	—
$U = \varphi_1 - \varphi_2$	Kuchlanish	U —volt (V)
$R = \rho \frac{l}{S}$	Qarshilik	—hajmiy qarshilik ($\Omega \cdot m$), l —uzunlik (m), S —kesim yuzasi (m^2)
$I = \frac{U}{R}$	Om qonuni	—
$j = \frac{I}{S} = \frac{1}{\rho} E$	Tok zichligi	j —A/ m^2

8.2 Qarshiliklarni ulash

Formula	Tavsif	Belgilar
$R = R_1 + R_2 + R_3 + \dots$	Ketma-ket ulash	Qarshiliklar qo' shiladi
$\frac{1}{R} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$	Parallel ulash	Teskari qarshiliklar qo' shiladi
$R = \frac{R_1 R_2}{R_1 + R_2}$	Ikki parallel qarshilik	Maxsus formula

8.3 Ish, quvvat va KPD

Formula	Tavsif	Belgilar
$P = UI = I^2 R = \frac{U^2}{R}$	Elektr quvvat	P —vatt (Vt)

Table 30 – continued

Formula	Tavsif	Belgilar
$A = UIt = I^2Rt = \frac{U^2}{R}t$	Elektr ish	A –joule (J) yoki kVt·soat
$Q = I^2Rt$	Joul-Lents qonuni	Issiqlik miqdori (J)
$\eta = \frac{P_{foydali}}{P_{umumiy}} \cdot 100\%$	Elektr zanjiri KPD	—

8.4 Tok manbai

Formula	Tavsif	Belgilar
$I = \frac{\mathcal{E}}{R + r}$	To' liq zanjir Om qonuni	—EMK (V), r —ichki qarshilik (Ω)
$\mathcal{E} = U + Ir$	Kuchlanish va EMK	U —tashqi kuchlanish
$U = \mathcal{E} - Ir$	Tashqi kuchlanish	Bo' sh o' tishda U =
$I_{qisqa} = \frac{\mathcal{E}}{r}$	Qisqa tutashuv toki	R = 0 bo' lganda

8.5 Kirchgof qonunlari

Formula	Tavsif	Belgilar
$\sum I_k = 0$	1-qonun (tugun)	Tugunga kirgan tok = chiqqan tok
$\sum \mathcal{E}_k = \sum I_k R_k$	2-qonun (kontur)	Zanjir bo' ylab EMK yig' indisi = kuchlanish yig' indisi

9. ELEKTROMAGNIT INDUKSIYA

9.1 Magnit maydon

Formula	Tavsif	Belgilar
$B = \frac{F_{max}}{I \cdot l}$	Magnit induksiyasi	B –teslas (Tl)
$B = \frac{\mu_0 I}{2\pi r}$	To'g'ri tok magnit maydoni	$= 4 \times 10^{-7}$ Gn/m –magnit doimiysi
$B = \mu_0 \mu n I$	Solenoid magnit maydoni	n –o'rma zichligi (o'rma/m), μ – magnit o'tkazuvchanlik
$\Phi = BS \cos \alpha$	Magnit oqimi	Φ –veber (Vb)

9.2 Kuchlar

Formula	Tavsif	Belgilar
$F = BIl \sin \alpha$	Amper kuchi	Tok o'tkazuvchi sim magnit maydonda
$F = qvB \sin \alpha$	Lorens kuchi	Harakatlanuvchi zaryad magnit maydonda
$M = BIS \sin \alpha$	Ramka momenti	Tok o'tkazuvchi ramka

9.3 Elektromagnit induksiya

Formula	Tavsif	Belgilar
$\mathcal{E}_i = -\frac{\Delta \Phi}{\Delta t}$	Faradey qonuni	Induksiya EMK
$\mathcal{E}_i = Blv \sin \alpha$	Harakatdan EMK	Sim maydon ichida harakatlansa

Table 35 – continued

Formula	Tavsif	Belgilar
$\mathcal{E}_s = -L \frac{\Delta I}{\Delta t}$	O' z-o' ziga induksiya	L –induktivlik (Gn)
$W = \frac{LI^2}{2}$	Induktivlik energiyasi	—
$L = \mu_0 \mu n^2 V$	Solenoid induktivligi	V –solenoid hajmi (m ³)

9.4 O'zgaruvchan tok

Formula	Tavsif	Belgilar
$I = I_0 \sin(\omega t)$	O' zgaruvchan tok	I –amplituda (A), $\omega = 2\pi f$
$U = U_0 \sin(\omega t)$	O' zgaruvchan kuchlanish	—
$I_{eff} = \frac{I_0}{\sqrt{2}}$	Samo' tariz (RMS)	Effektiv qiymat
$U_{eff} = \frac{U_0}{\sqrt{2}}$	Kuchlanish effektiv qiymati	—
$X_C = \frac{1}{\omega C}$	Sig' imiy reaktiv qarshilik	Xc –oms (Ω)
$X_L = \omega L$	Induktiv reaktiv qarshilik	XL –oms (Ω)
$Z = \sqrt{R^2 + (X_L - X_C)^2}$	To' la qarshilik (impedans)	Z –oms (Ω)
$\tan \varphi = \frac{X_L - X_C}{R}$	Silsila fazavish siljishi	—burchak

Table 36 – continued

Formula	Tavsif	Belgilar
$I = \frac{U}{Z}$	O' zgaruvchan tok Om qonuni	—
$f_0 = \frac{1}{2\pi\sqrt{LC}}$	Rezonans chastotasi	—

9.5 Transformator

Formula	Tavsif	Belgilar
$\frac{U_1}{U_2} = \frac{n_1}{n_2} = \frac{I_2}{I_1}$	Transformator tenglamasi	n —o' rma soni
$\eta = \frac{P_2}{P_1} \cdot 100\%$	Transformator KPD	—

10. OPTIKA

10.1 Geometrik optika

Formula	Tavsif	Belgilar
$n = \frac{c}{v}$	Sinish ko' rsatkichi	n —sinish ko' rsatkichi, c —yorug' lik tezligi (3×10^8 m/s)
$n_1 \sin \alpha = n_2 \sin \beta$	Snuell qonuni	—tushish burchagi, —sinish burchagi
$\sin \alpha_{crit} = \frac{n_2}{n_1} \quad (n_1 > n_2)$	To' liq ichki akslanish	α_{crit} —kritik burchak

10.2 Aks ettirish

Formula	Tavsif	Belgilar
$\alpha = \beta$	Aks ettirish qonuni	Tushish burchagi = aks ettirish burchagi
$\frac{1}{F} = \frac{1}{d} + \frac{1}{f}$	Linza (ko' zgu) formulasi	F —fokus masofasi (m), d —buyum masofasi, f —tasvir masofasi
$\Gamma = \frac{H}{h} = \frac{f}{d}$	Kattalashtirish	H —tasvir balandligi, h —buyum balandligi
$D = \frac{1}{F}$	Optik kuchi	D —dioptriya (dptr)

10.3 Fizik optika

Formula	Tavsif	Belgilar
$\Delta x = \frac{\lambda L}{d}$	Ikki yoriq interferensiyasi	Δx —oraliq (m), L —ekran masofasi, d —yoriq orasi
$d \sin \varphi = k \lambda$	Difraksion panjara	$k = 0, \pm 1, \pm 2 \dots$ —tartib raqami
$l \sin \varphi = k \lambda$	Faungofer difraktsiyasi	l —yoriq kengligi
$L = \frac{\lambda}{\sin \alpha} \approx \frac{\lambda}{\alpha}$	Ko' rish chegarasi	L —eng kichik ayrilishi mumkin bo'lgan masofa

10.4 Yorug'likning kvant xususiyatlari

Formula	Tavsif	Belgilar
$\varepsilon = hf = \frac{hc}{\lambda}$	Foton energiyasi	$h = 6.63 \times 10^{-34}$ J·s —Plank doimiyasi
$p = \frac{h}{\lambda} = \frac{\varepsilon}{c}$	Foton impulsi	p —impuls (kg·m/s)

Table 41 – continued

Formula	Tavsif	Belgilar
$h\nu = A_{chiq} + E_{k,max}$	Fotoeffekt (Einstein)	A_{chiq} —chiqish ishi (J), $E_{k,max}$ —maksimal kinetik energiya
$eU_{to'xt} = E_{k,max} = h\nu - A_{chiq}$	To'xtatish potentsiali	$U_{to'xt}$ —to'xtatish potentsiali (V)
$\nu_0 = \frac{A_{chiq}}{h}$	Chegaraviy chastota	—threshold chastota (Hz)

11. ATOM VA YADRO FIZIKASI

11.1 Atom strukturasi (Bor modeli)

Formula	Tavsif	Belgilar
$r_n = n^2 \cdot r_1 = n^2 \cdot \frac{\hbar^2}{kme^2}$	n-darajadgi radius	$r_1 = 0.53 \times 10^{-10} \text{ m} = 0.53 \text{ \AA}$ —1-sayt radius
$r_n = n^2 \frac{r_1}{Z}$	Vodorodga o'xshash atomlar	Z —atom raqami
$v_n = \frac{v_1}{n}$	Elektron tezligi	$v_1 = 2.19 \times 10^6 \text{ m/s}$ —1-sayt tezlik
$E_n = -\frac{13.6 \cdot Z^2}{n^2} \text{ eV}$	Energiya darajalari	Vodorod atomi uchun
$E_n = \frac{E_1}{n^2} \quad (E_1 = -13.6 \text{ eV})$	Vodorod energiyalari	—
$\frac{1}{\lambda} = R \left(\frac{1}{n^2} - \frac{1}{m^2} \right)$	Balmer formulasi	$R = 1.097 \times 10^7 \text{ m}^{-1}$ —Ridberg doimiyasi

Table 42 – continued

Formula	Tavsif	Belgilar
$L = mvr = n\hbar = n\frac{h}{2\pi}$	Moment impuls kvantlanishi	$n = 1, 2, 3 \dots$ —bosh kvant soni

11.2 de Broy to'liqlari

Formula	Tavsif	Belgilar
$\lambda = \frac{h}{p} = \frac{h}{mv}$	de Broy to'liq uzunligi	p —zarracha impulsi
$\lambda = \frac{h}{\sqrt{2mE_k}} = \frac{h}{\sqrt{2meU}}$	Zaryadlangan zarracha	U —tezlatuvchi kuchlanish (V)
$\lambda = \frac{1.226}{\sqrt{U(\text{V})}} \text{ nm}$	Elektron (qulay formula)	U kV da bo'lsa: $\lambda = 1.226/\sqrt{U}$ (pm)

11.3 Yadro fizikasi

Formula	Tavsif	Belgilar
$E = \Delta m \cdot c^2$	Massa-energiya ekvivalentligi	$c = 3 \times 10^8$ m/s —yorug'lik tezligi
$\Delta m = Zm_p + Nm_n - M_{yadro}$	Massa defekti	Z —protonlar soni, N —neytronlar soni
$E_{bog} = \Delta m \cdot c^2$	Bog'lanish energiyasi	—
$\varepsilon = \frac{E_{bog}}{A}$	Nuklon uchun bog'lanish energiyasi	$A = Z + N$ —massa soni

11.4 Radioaktiv parchalanish

Formula	Tavsif	Belgilar
$N = N_0 \cdot 2^{-t/T_{1/2}}$	Parchalanish qonuni	N —boshlang' ich yadrolar soni
$N = N_0 e^{-\lambda t}$	Parchalanish (eksponensial)	—parchalanish doimiysi
$T_{1/2} = \frac{\ln 2}{\lambda} = \frac{0.693}{\lambda}$	Yarim parchalanish davri	$T^{1/2}$ —yarim parchalanish davri
$A = \lambda N = \frac{\ln 2}{T_{1/2}} N$	Aktivlik	A —bekerel (Bq)

11.5 Radioaktiv parchalanish turlari

Turi	Tenglama	O' zgarish
Alfa ()	${}_Z^AX \rightarrow {}_{Z-2}^{A-4}Y + {}_2^4\text{He}$	$Z \rightarrow Z-2, A \rightarrow A-4$
Beta ()	${}_Z^AX \rightarrow {}_{Z+1}^AY + {}_{-1}^0e + \tilde{\nu}$	$Z \rightarrow Z+1, A = \text{const}$
Beta ()	${}_Z^AX \rightarrow {}_{Z-1}^AY + {}_{+1}^0e + \nu$	$Z \rightarrow Z-1, A = \text{const}$
Gamma ()	${}_Z^AX^* \rightarrow {}_Z^AX + \gamma$	$Z = \text{const}, A = \text{const}$ (energiya ajraladi)

12. MAXSUS NISBIYLIK NAZARIYASI

Formula	Tavsif	Belgilar
$\gamma = \frac{1}{\sqrt{1 - \frac{v^2}{c^2}}}$	Lorens faktor	(gamma) —nisbiylik omili

Table 47 – continued

Formula	Tavsif	Belgilar
$l = l_0 \sqrt{1 - \frac{v^2}{c^2}} = \frac{l_0}{\gamma}$	Uzunlik qisqarishi	l — tinchlikdagi uzunlik
$\Delta t = \frac{\Delta t_0}{\sqrt{1 - \frac{v^2}{c^2}}} = \gamma \Delta t_0$	Vaqt cho' zilish	Δt — o' z vaqt (proper time)
$m = \frac{m_0}{\sqrt{1 - \frac{v^2}{c^2}}} = \gamma m_0$	Nisbiylik massasi	m — tinchlik massasi (rest mass)
$E = mc^2 = \gamma m_0 c^2$	Jami energiya	E — jami energiya (J)
$E = E_0 + E_k = m_0 c^2 + E_k$	Energiya (ikki qismda)	$E = m c^2$ — tinchlik energiyasi
$E_k = mc^2 - m_0 c^2 = m_0 c^2 (\gamma - 1)$	Kinetik energiya (nisbiylik)	—
$p = mv = \gamma m_0 v$	Nisbiylik impulsi	—
$E^2 = p^2 c^2 + m_0^2 c^4$	Energiya-impuls bog' lanishi	Massasiz zarrachalar uchun $E = pc$

QO'SHIMCHA: SI TIZIMI ASOSIY BIRLIKLARI

Kattalik	Birlik	Belgisi
Uzunlik	metr	m
Massa	kilogramm	kg
Vaqt	soniya	s
Tok kuchi	amper	A
Temperatura	kelvin	K
Modd miqdori	mol	mol
Yorug' lik kuchi	kandela	cd

Table 48 – continued

Kattalik	Birlik	Belgisi
Asosiy fizik doimiylar		
Doimiy	Qiymat	
Yorug'lik tezligi (c)	3×10^8 m/s	
Plank doimiyasi (h)	6.63×10^{-34} J·s	
Gravitatsion doimiy (G)	6.67×10^{-11} N·m ² /kg ²	
Gaz doimiysi (R)	8.31 J/(mol·K)	
Boltsman doimiyasi (k)	1.38×10^{-23} J/K	
Avogadro soni (N)	6.02×10^{23} mol ⁻¹	
Elektron zaryadi (e)	1.6×10^{-19} C	
Elektron massasi (m_e)	9.11×10^{-31} kg	
Proton massasi (m_p)	1.673×10^{-27} kg	
Neytron massasi (m_n)	1.675×10^{-27} kg	
Vakuum dielektrik o'tkazuvchanligi (ϵ_0)	8.85×10^{-12} F/m	
Vakuum magnit o'tkazuvchanligi (μ_0)	4×10^{-7} Gn/m	

PhysicsLab — Formulalar to'plami to'liq. Har bir formula KaTeX formatida va uni saytda ishlatish uchun tayyor. Foydalanuvchilar qidiruv, sevimlilar va nusxa olish funksiyalari orqali osongina foydalanishlari mumkin.